When Does Non-Myopic Bayesian Experimental Design Work with Language Models?

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Abstract

Sequential Bayesian experimental design (BED) offers a principled way for language-model agents to ask questions, update beliefs, and plan future information gathering. Yet deeper planning can optimize errors in an LLM-derived probabilistic model rather than useful information. We audit this failure across an exact constrained-location test, semantic 20 Questions, and two external interactive benchmarks: Paprika troubleshooting and MediQ clinical reasoning. Depth-two planning succeeds in the exact test, while one-step EIG substantially beats prompting in 20 Questions. We then test StrategyEIG: an LLM proposes short contingent policies and an exact Bayesian verifier selects their root. Across four preregistered Rock Diagnosis maps with three to eleven rocks, StrategyEIG improves entropy AUC over the same LLM roots scored myopically by 0.475–0.941, exhaustive one-step width by 0.452–0.939, and matched random strategies by 0.362–0.887; all twelve paired 95% intervals exclude zero, and truth-log-posterior AUC agrees. Both the eight- and eleven-rock results replicate across Gemma 4 26B A4B and GPT-5.4 Mini under shared root slots. The selected policies move to enable accurate future checks, whereas both myopic controls never move. Negative location, Paprika, and MediQ audits identify the boundary: depth needs both a deployment-matched generative model and verified non-greedy structure. The result supports an LLM-Modulo architecture for sequential BED—semantic policy generation coupled to exact external verification—rather than unverified LLM simulation alone.

1 Introduction

Interactive agents must often acquire information before committing to an answer or action. A clinical agent asks about symptoms; a support agent diagnoses a fault; a guessing agent narrows a large semantic space. Direct prompting is an unreliable solution: MediQ reports that asking models to interact can reduce accuracy relative to answering from the initial context [Li et al., 2024]. Bayesian experimental design instead selects a query by its expected information gain (EIG) about a target variable. BED-LLM makes this practical in semantic spaces by using an LLM to propose hypotheses and provide response likelihoods [Choudhury et al., 2026].

The natural next step is non-myopic BED: score a query by both its immediate EIG and the information available to a future adaptive query. Recent methods simulate conversation trees for this purpose [Hu et al., 2024, Arnould et al., 2026]. In principle this can value a prerequisite question with little immediate gain. In practice every additional branch invokes the same uncertain model, changes the belief on which later calls condition, and maximizes over more noisy continuation scores. A larger tree can therefore make a confidently worse decision.

We investigate this distinction using a validation-first sequence of experiments. An exact constrained-location environment verifies the recursion. Rock Diagnosis then supplies the missing combination of an exact simulator, enabling movement, LLM-generated branch policies, and external exact verification. Semantic 20 Questions verifies one-step transfer but does not establish an incremental horizon benefit. Natural location experiments show non-monotonic depth under LLM-generated strategies. Finally, Paprika and MediQ expose different failures on external benchmarks: task utility and endpoint attribution in Paprika, and latent-state sufficiency plus record sparsity in MediQ.

Our contributions are:

1. StrategyEIG, an LLM-Modulo sequential BED architecture in which an LLM proposes explicit contingent policies and an exact Bayesian component verifies their total information gain [Kambhampati et al., 2024];

2. preregistered paired evidence that both the LLM proposal prior and non-myopic verification are load-bearing in a depth-favorable Rock geometry;

3. a cross-environment audit that separates correctness of the planning recursion from validity of the probabilistic interaction model;

4. two external-benchmark autopsies with invalid outcomes quarantined rather than repurposed as method evidence; and
5. a concrete validation contract for future non-myopic LLM BED: target and latent sufficiency, calibrated likelihoods, branch/deployment equivalence, a verified greedy gap, and ranking fidelity before policy evaluation.

2 The Probabilistic Contract

Let $h_t = ((x_1, y_1), \ldots, (x_t, y_t))$ be the interaction history and θ the finite target decoded at the endpoint. A generative state z may contain additional information needed to predict responses, with $\theta = g(z)$. The relevant model is

$$p_t(\theta, z)\, p(y \mid z, x, h_t), \qquad p(y \mid \theta, x, h_t) = \sum_z p(y \mid z, x, h_t) p_t(z \mid \theta).$$

When θ itself is sufficient, as for a hidden animal answering a property question, z can be omitted. When θ is merely a decision label, such as "treat hypoperfusion first," it does not determine a patient's glucose or history and cannot serve as the response-generating state.

For a coherent model, one-step EIG is

$$I_t(x) = \sum_{\theta, y} p_t(\theta) p(y \mid \theta, x, h_t) \log \frac{p(y \mid \theta, x, h_t)}{\sum_{\theta'} p_t(\theta') p(y \mid \theta', x, h_t)}.$$

Our exact depth-two score is

$$Q_2(x) = I_t(x) + \mathbb{E}_{y \mid x, h_t} \left[\max_{x' \in \mathcal{X}(h_t, x, y)} I_{t+1}(x') \right].$$

It branches over every positive-mass response, performs a Bayesian update on the fixed target support, generates branch-specific follow-ups, and adds the best second-step EIG. This is mathematically appropriate for the model supplied to it. It cannot determine whether that model describes deployment.

Endpoint target $\theta \to$ sufficient state $z \to$ calibrated $p(y \mid z, x, h) \to$ matched simulated/deployed update $\to$ verified greedy gap $\to$ rankable depth increment $\to$ endpoint comparison

Figure 1: The validation chain for non-myopic LLM BED. Failure of any upstream link invalidates evidence from downstream policy performance.

Two requirements are therefore distinct. *Model validity* asks whether the target, latent state, likelihood, update, and simulated branches form one process that matches deployment. *Planning value* asks whether the task contains complementarity or gating such that a lower-immediate-EIG query enables a more informative continuation. If information has adaptive diminishing returns, greedy policies can already be competitive [Golovin and Krause, 2011]. Depth needs a demonstrated gap, not merely a multi-turn interface.

3 Audit Design and Evidence Rules

We aggregate only preregistered or reconstructable paired evidence. Exact location tests use the known source prior and signal likelihood. The animals audit reconstructs every 20-round binary trial and reports deterministic bootstrap intervals. Paprika and MediQ use hash-pinned official releases and strict transcript audits. Failed parsing, unsupported simulator facts, and endpoint contradictions fail closed.

Most importantly, we distinguish *policy evidence* from *diagnostic evidence*. A Paprika run whose customer reports the private correct remedy as failing cannot compare policies, even if aggregate resolution numbers exist. A MediQ run with an invalid likelihood cannot establish EIG value, even if its queries are fluent. Such runs may localize a broken link in Figure 1; they do not support an accuracy claim.

Setting	Generative model	Greedy gap	Key result	Evidence status
Exact constrained location	Exact	Explicit delayed trap	RMSE: d2 .153, greedy .561	Positive control
Rock Diagnosis	Exact finite state	Enabling movement	Strategy d2 beats d1, width, random	Paired positive
Animals 20Q	Concrete target, binary replies	Unverified	d1 AUC .743, think-naive .465	One-step positive
Natural location	Exact signal, LLM strategies	Weak/variable	d1/d3/d5 non-monotonic	Horizon negative
Paprika	Insufficient state/task utility	Assumed, not enforced	Endpoint contradiction	Diagnostic only
MediQ iMEDQA	A–D often not generative	Channel too sparse	8/30 replies available	Gate failure

Table 1: The same recursion behaves differently as the probabilistic and task contracts weaken. "d1" and "d2" denote scoring horizon.

4 Results Across Environments

Exact lookahead can work. We first constructed a branch-decoy source prior with a local-bump observation model and movement-constrained queries. The first query can reveal which region to enter, while movement limits make that choice affect later measurements. Over 120 paired trials, exact depth two reduced final RMSE from 0.5606 for greedy EIG to 0.1529, a mean paired difference of -0.4077. This is a harness result, not an external-benchmark claim. It shows that the implementation and objective can exploit a genuine delayed-information structure when model error is absent.

One-step semantic BED transfers; depth does not yet. On animals 20 Questions, depth-one EIG achieved accuracy AUC 0.743 versus 0.465 for a thinking naive policy and 0.458 for non-thinking naive over 40 paired trials. The AUC gain over thinking naive was 0.278 with 95% bootstrap interval $[0.191, 0.365]$. This is the stable positive BED result.

The apparent aggregate depth-two AUC was 0.770, but the full depth-one and depth-two streams were not paired. On the only exact ten-trial seed block, depth two minus depth one was -0.040 with interval $[-0.105, 0.025]$ and 3/1/6 wins/ties/losses. Depth three was worse than depth two across 40 paired trials: -0.126 $[-0.224, -0.035]$. Thus these data support semantic EIG, not non-myopic improvement.

Natural location experiments reinforce this distinction. In a 30-trial movement-constrained study using exact physical likelihoods but LLM-generated strategies, final RMSE was 0.412 for greedy EIG and 0.929, 1.127, and 0.542 for strategy horizons 1, 3, and 5. Without the movement constraint all horizons were effectively flat. More planning did not yield monotonic behavior, and the naive policy remained competitive.

LLM proposals plus exact lookahead work in enabling geometry. The negative natural-location sweep did not test the architecture where depth itself was useful: a later exact width gate showed that its smooth sensing geometry favored one-step action coverage. We therefore moved to finite Rock Diagnosis. Movement has exactly zero immediate EIG, but changes distance-dependent sensor accuracy. The LLM returned six explicit two-action branch policies per decision: three distinct movement roots followed by checks, and three direct-check roots. A finite exact posterior and exhaustive observation branching scored every policy; the LLM was never called inside rollout scoring.

We preregistered 30 paired eight-round trajectories on each of two paper maps, then scale extensions on the canonical eight-rock RockSample[7,8] geometry from Smith and Simmons [2004] and the standard eleven-rock RockSample[11,11] SARSOP instance. This is a diagnosis-only adaptation: it retains the latent rock vector, grid movement, and distance-dependent sensor, while excluding sampling, reward, and exit actions. Controls remove one component at a time: shared d1 scores the identical LLM roots myopically; width uses exhaustive d1 over all legal actions with matched exact-scorer units and one LLM ordering call; random strategy uses the same branch grammar and exact d2 verifier with random legal policies. Table 2 reports paired gains in mean posterior-entropy AUC; positive is better for StrategyEIG.

Map	Control	Entropy-AUC gain	95% CI	W/T/L
3–6	Shared-roots d1	.5205	[.4857, .5531]	30/0/0
3–6	Exhaustive d1 width	.5205	[.4855, .5537]	30/0/0
3–6	Random strategies	.3748	[.3229, .4267]	30/0/0
5–7	Shared-roots d1	.4751	[.3929, .5544]	30/0/0
5–7	Exhaustive d1 width	.4517	[.3677, .5310]	29/0/1
5–7	Random strategies	.3622	[.2714, .4482]	29/0/1
7–8	Shared-roots d1	.7502	[.7050, .7938]	30/0/0
7–8	Exhaustive d1 width	.7297	[.6874, .7724]	30/0/0
7–8	Random strategies	.6735	[.6251, .7212]	30/0/0
11–11	Shared-roots d1	.9412	[.8968, .9807]	30/0/0
11–11	Exhaustive d1 width	.9387	[.8952, .9773]	30/0/0
11–11	Random strategies	.8867	[.8305, .9385]	30/0/0

Table 2: Preregistered StrategyEIG confirmation. All twelve primary paired bootstrap intervals exclude zero. Truth-log-posterior AUC also favors StrategyEIG in all twelve comparisons, with all twelve intervals excluding zero.

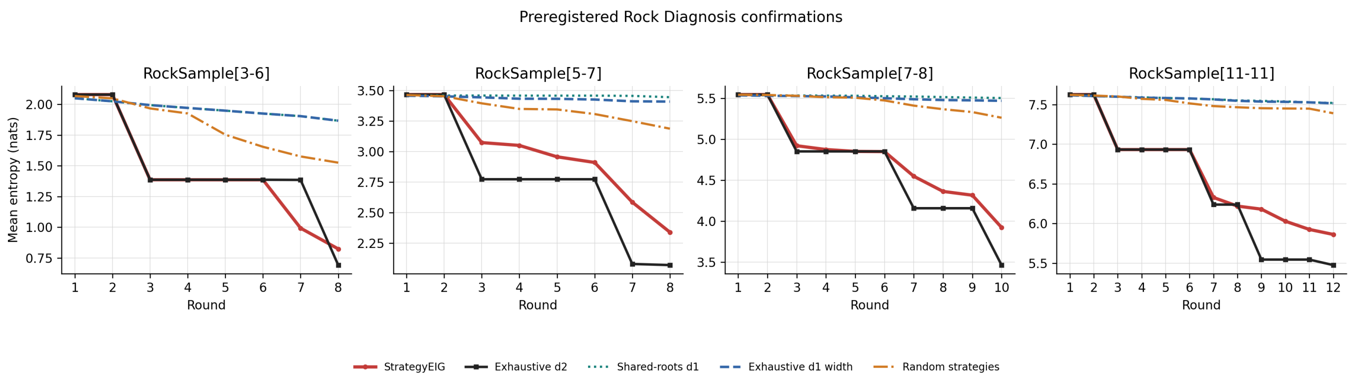

Figure 2: Mean posterior entropy over 30 paired trajectories per map. StrategyEIG and exhaustive d2 first spend zero-information actions on movement, then obtain large entropy drops from more accurate checks.

The mechanism is direct. StrategyEIG selects movement on 140/240 and 128/240 decisions on the smaller maps, 193/300 on RockSample[7,8], and 195/360 on RockSample[11,11], while shared d1 and exhaustive d1 width never move. Its proposals capture 90.6%, 86.1%, 85.8%, and 73.9% of exhaustive d2 value over depth-two rounds. Before the paid eight-rock scale extension, a 1,000-pair exact qualification found that exhaustive d2 reduced final entropy by 1.6775 nats relative to exhaustive d1, with 95% interval [1.6568, 1.6980]. Beating shared d1 shows that non-myopic scoring is load-bearing; beating matched random strategies shows that the LLM prior is load-bearing. The exhaustive d2 oracle remains better. A preregistered zero-call diagnostic closed each Gemma root over exact legal followups, raising mean exhaustive fraction from 85.8% to 99.6% and localizing most headroom to continuation choice. Optimal-root coverage was 242/270 (89.6%), one state below the frozen 90% paid-run gate, so we did not tune or run it.

Under fresh seed and machine-assigned movement-root slots, GPT-5.4 Mini also beat shared d1, exhaustive d1 width, and random policies: .5104 [.4312, .5863], .4928 [.4115, .5734], and .4130 [.3225, .4988], with 28/0/2 each; all truth-log intervals excluded zero. The sign is not Gemma-specific, although GPT's smaller gains and 66.3% exhaustive fraction leave proposal headroom.

To remove the remaining interface difference, we preregistered a second fresh-seed Gemma run under the identical movement-root slots. Its entropy-AUC gains were .7846 [.7370, .8296] against shared d1, .7729 [.7252, .8167] against exhaustive d1 width, and .7334 [.6813, .7835] against random strategies, with 30/0/0 wins/ties/losses in every comparison. All truth-log-AUC intervals again excluded zero. The run completed all 870 proposal cells without a rejection or resume and narrowed the gap to exhaustive d2 from .1303 in the semantic-prompt Gemma run to .0814. This same-interface replication removes prompt format as an explanation for the cross-family positive result. Effect-size differences across prompts and models remain descriptive because the runs use different seeds; they are not causal interface or model-superiority tests.

Finally, a 500-pair exact gate on RockSample[11,11] found that exhaustive d2 beat exhaustive d1 by 1.1042 entropy-AUC nats [1.1015, 1.1068]. Four independent Gemma seeds each passed all six entropy/truth gates. Across the three fresh robustness seeds, pooled entropy-AUC gains were .9470 [.9253, .9671], .9440 [.9228, .9637], and .8727 [.8359, .9073], with 90/0/0 each. GPT-5.4 Mini also passed all six, although its .5459 oracle gap shows model-dependent proposal quality. This extends the result to 2,048 latent states with zero rollout LLM calls. Exact closure raised LLM roots from 73.9% to 96.9% of oracle value, but random roots reached 96.7%, localizing the LLM advantage to continuation choice; the paid gate failed. Across maps, K6 uses 13.7–14.2 scorer nodes/decision versus exhaustive d2's 64.3–354.2 (4.7×–24.9×); these are tree-width, not wall-clock, ratios. A fresh 11-rock width sweep passed all 18 gates: K4 beat K2 by .5655 entropy-AUC nats [.4974, .6418], while K6 minus K4 was .0102 [-.0156, .0337]. K4 therefore matched K6 endpoint quality with 9.49 instead of 14.24 scorer nodes/decision.

More receding-horizon depth is not automatically better, even with the exact simulator. In a preregistered 500-pair gate, terminal-EIG d3 lost to d2 by .2044 entropy-AUC nats, with 95% interval [.1998, .2090] in favor of d2 and 0/0/500 d3 wins/ties/losses; truth-log AUC agreed. At the first divergence, d3 takes a noisy check because its local value includes a promised later move and perfect check. Replanning restores the three-step window, so it repeats the check and postpones the continuation. A fresh endpoint-aligned repair weighted earlier gains by their remaining AUC exposure. It made d3 beat same-utility d2 by .2744 [.2689, .2799], 500/0/0, but exactly reproduced the prior best terminal-EIG d2 trajectory and score. Alignment therefore removes the rolling-horizon reversal without establishing a strict deeper oracle gain; we did not spend on an h3 LLM run.

Paprika fails before a horizon claim. Paprika troubleshooting mixes information-gathering questions with corrective actions that change and may terminate the world [Tajwar et al., 2025]. Pure EIG values discrimination among cause/remedy strings, not the probability of resolution minus action cost. A remedy that resolves under every hypothesis can have zero EIG, while a diagnostic that identifies the cause but cannot resolve the task can score highly.

The implementation also used fixed support and synthetic outcome labels in lookahead while deployment refreshed support, observed free text, and could terminate. A ten-task full-depth diagnostic used 70,004 requests versus 3,365 for one-step EIG and resolved 2 versus 3 tasks. Those counts are not policy evidence: later manual audit found a customer response claiming that the exact private remedy had failed. A separate 50-task arbitration evaluation was also quarantined by endpoint review. The defensible result is the failure mechanism, not which arm won an invalid simulator contest.

MediQ validates the interface but rejects the joint model. MediQ's official FactSelect patient returns only atomic record facts that answer the question and otherwise abstains [Li et al., 2024]. After several failed closed repairs, the final five-case environment gate achieved clean canonical Yes/No/unavailable questions, verbatim grounding, relevance, and semantic deduplication on every selected interaction. This isolates the remaining issue to probability modeling.

The first scorer treated each A–D option as a complete patient world. Across ten selected turns, predicted EIG averaged 0.140 nats but realized true-label log-probability gain averaged -0.290 nats; the realized outcome was less likely under the true label than under the predictive mixture on 6/10 turns. A repair made record missingness label-independent and scored only Yes versus No under each option. Missingness became exactly posterior-neutral, yet available outcomes still had mean true-label log gain -0.118 nats.

A final data-estimation construction elicited the response marginal and hypothetical A–D posteriors, then projected their joint table to exact local marginals. Before scoring it, a frozen held-out bank required at least 15 available responses. All 30 questions were canonical, unique, grounded, and relevant, but only 8 were answerable; 22 facts were absent from the static records. The scorer was therefore not run. At observed answerability $a = 8/30$, an unavailable-neutral binary channel has the loose ceiling

$$I(\theta; Y \mid x) \leq a \log 2 = 0.1848 \text{ nats per query},$$

before accounting for diagnostic ambiguity. There is little signal on which a depth-two advantage could operate.

We then ran a fresh, preregistered iCRAFT-MD diagnosis-only profile gate. The official scorer uses a four-way diagnosis index and the records are denser, so we assigned three concrete counterfactual patient profiles to each diagnosis, held record availability label-independent, and froze support before any branch. The one-case micro-smoke passed: it produced all 12 profiles, two candidates, and canonical profile likelihood tables for $0.0027. The original iCRAFT profile-support gate failed before likelihood when the 26B scaffold exhausted bounded profile construction on calibration. One separately authorized retry used a frontier model to author profiles only, with 26B retaining every other role. It reached calibration but only 14 of 48 grounded FactSelect outcomes were available, below the preregistered minimum of 24. This terminal channel-sparsity failure closes the path before structural, ranking, or policy evaluation; we made no prompt repair or rerun.

5 Why Horizon Magnifies Model Error

Let an estimated continuation score be $\widehat{V}_j = V_j + \epsilon_j$ for branch candidate j. Depth two uses $\max_j \widehat{V}_j$, so even zero-mean errors produce positive selection bias. Increasing branch width or horizon adds more opportunities for an optimistic likelihood, posterior, or proposal to win. Root likelihood error also changes branch mass and the posterior context passed to every downstream call. More Monte Carlo rollouts reduce variance conditional on this model; they do not reduce its bias. Bayesian design is particularly exposed to misspecification because the same model both predicts data and decides which data to collect [Catanach and Das, 2023].

BED-LLM reaches strong one-step results under cleaner conditions: a concrete entity or user profile, a small response space, a prior-likelihood construction, and diverse hypotheses retained with uniform mass after history filtering [Choudhury et al., 2026]. It explicitly reports raw in-context beliefs as overconfident and finds its data-estimation ablation weaker than prior-likelihood BED. Our MediQ belief similarly reached mean top probability 0.867 after round one at only 50% accuracy.

The Rock result instead follows the LLM-Modulo pattern: the model supplies a small, semantically informed policy set, while a sound external component checks grammar, legality, Bayesian updates, and total information gain [Kambhampati et al., 2024]. The comparison to random policies shows that the generator contributes useful search bias; the comparison to the same roots under d1 shows that external lookahead contributes the decision rule. Neither component alone explains the gain.

UoT reports improvements with simulation depth, but operates on explicit possibility sets and repeatedly partitions the same support; its diagnosis tasks contain only 5 or 15 diseases [Hu et al., 2024]. CA-BED uses finite mutually exclusive hypotheses, soft LLM likelihoods, binary questions, and depth two on structured entity deduction [Arnould et al., 2026]. It improves over direct prompting, but does not report a causal depth-one versus depth-two ablation; its categorical-answer extension hurts the harder Detective benchmark. These papers show that structured planning can help, not that horizon is robust to an insufficient latent or incompatible answer categories.

Deep adaptive design and RL-BED optimize total sequential utility through amortized policies or reinforcement learning [Foster et al., 2021, Blau et al., 2022]. Both require trajectories from a meaningful simulator. Training on the current MediQ surrogate would amortize its annotation and calibration artifacts rather than recover a missing patient state.

6 A Validation-First Path Forward

Future depth studies should cross the following gates in order:

1. **Target and state:** the endpoint must decode θ, while a latent z must contain everything required to generate responses.
2. **Prior:** held-out log loss and Brier score must beat uniform, with no high-confidence wrong collapse.
3. **Likelihood:** available observations must improve mean true-target log probability; missingness must be neutral unless its semantics are genuinely target-dependent.
4. **Branch equivalence:** simulated and deployed histories must yield the same support, probabilities, and stopping behavior.
5. **Structural gap:** an exact or high-fidelity oracle must show depth two beating greedy under the benchmark's native action and budget rules.
6. **Ranking fidelity:** estimated incremental lookahead value must rank oracle incremental value under shared candidates and common random numbers.
7. **Policy test:** only then compare depth one and two on paired cases, with identical root candidates, simulator seeds, and stopping rules.

Within MediQ, the official iCRAFT-MD split offered a fresh diagnosis-only target: all 140 targets are diagnoses and records have a mean 14.82 atomic facts (median 14, minimum 9), compared with 11.20 (median 10, minimum 2) in usable iMEDQA. We preregistered and tested the natural repair—several concrete counterfactual patient profiles z per diagnosis, profile-conditioned replies, label-independent availability, and EIG about $\theta = g(z)$. A stronger author could construct the fixed support, but only 14/48 held-out FactSelect outcomes were answerable, short of the frozen 24-outcome requirement. This is an upstream channel/model-validity failure, not evidence that a later policy would fail. It nevertheless closes this un-tuned path: we do not promote iCRAFT as an unevaluated escape hatch.

7 Limitations and Conclusion

This is a validation study and a positive structured-benchmark result, not yet a positive external-benchmark claim for non-myopic LLM BED. Rock Diagnosis has an exact finite simulator and deliberately visible enabling structure; even the canonical RockSample[7,8] and standard RockSample[11,11] extensions are diagnosis-only adaptations without the benchmark's reward, sampling, or exit actions. Rock Diagnosis does not test robustness to learned likelihoods or open-ended responses. Animals establishes one-step transfer, but its full depth-one and depth-two streams are only partially paired and have weaker code provenance than the newer audits. Natural location results use one model family and environment. Paprika policy counts are endpoint-invalid, and MediQ stops before a calibrated policy comparison. We do not claim that non-myopic BED cannot work; we report that the preregistered iCRAFT profile construction and its one stronger-author retry did not pass the held-out calibration gate.

The audits also cannot cleanly separate model capacity from representation: a larger model may estimate a better likelihood, but cannot make an exam-treatment label a sufficient patient state or reveal facts absent from a static record. OpenRouter provider nondeterminism, finite generated supports, different fresh seeds across the Rock replications, and post-hoc synthesis across heterogeneous experiments remain limitations.

The evidence supports a constructive conclusion. With a coherent simulator and an enabling-information gap, LLM proposals plus exact rollout-EIG verification produce a large, paired, non-myopic gain; both generation and verification matter. One-step semantic EIG can also improve information gathering. But depth is not a substitute for a valid joint model, deployment-matched branches, or a task in which greedy information gain is genuinely suboptimal. These are empirical gates, not implementation details. Testing them before scaling horizon is both cheaper and scientifically stronger than explaining another non-monotonic depth sweep after the fact.

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